System Validation Report

Vehicle Parking System | Team 10 | Software Engineering Mini Project

Covers: end-to-end customer and admin workflows run against the running application

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Build tested	main branch, Team10-VPS/vehicle_parking_system (zip snapshot)
Environment	Python 3.12, Flask 3, Flask-SQLAlchemy, SQLite, pytest 9, coverage 7

1. Approach

The app was seeded with seed.py and started with the Flask server on localhost:5055 with CSRF protection switched on (same as normal use, unlike the unit test config). A script (docs/system_validation_script.py) then drove the app over HTTP like a browser would: it loads each page, reads the CSRF token from the form and submits it, using separate sessions for two customers and the admin. This validates the full flow of page, route, database and session, which the Test Plan lists as system testing.

2. Summary

Scenarios executed	20
Passed	20
Failed	0

3. Scenario results

ID	Scenario	Expected	Actual	Status
SV-01	Customer registers a new account	Redirected to login page after registration	HTTP 200, final URL /login	Pass
SV-02	Customer logs in	Dashboard with slot list is shown	final URL /dashboard, floors shown: 3	Pass
SV-03	Customer opens booking form from an available slot	Booking form for the chosen slot is shown	HTTP 200	Pass
SV-04	Customer registers a vehicle	Vehicle appears on My Vehicles page	HTTP 302	Pass
SV-05	Customer books slot and is checked in	Redirect to history; active booking with check-in time	HTTP 302 to /history	Pass
SV-06	Booked slot shows as occupied on dashboard	Slot no longer offered as an available link	link for booked slot absent	Pass
SV-07	Second customer tries to book the occupied slot	Booking rejected (409)	HTTP 409	Pass
SV-08	Other customer tries to check out someone else's booking	403 Forbidden	HTTP 403	Pass
SV-09	Owner checks out	Booking completed and slot available again	HTTP 302; slot link back on dashboard: True	Pass
SV-10	Repeat check-out of a completed booking	409 Conflict	HTTP 409	Pass
SV-11	Customer opens admin page	403 Forbidden	HTTP 403	Pass
SV-12	Anonymous user opens admin page	Redirect to login	HTTP 302 to /login	Pass
SV-13	Admin adds a slot	New slot SV-99 listed	HTTP 200	Pass
SV-14	Admin adds a duplicate slot	Rejected with a message, no second slot	HTTP 200; SV-99 rows: 1	Pass
SV-15	Admin views active bookings	Active booking listed with force check-out option	force check-out buttons: 1	Pass

ID	Scenario	Expected	Actual	Status
SV-16	Admin force check-out	Booking completed and removed from active list	HTTP 302	Pass
SV-17	Admin removes an available slot	Slot removed	HTTP 302; SV-99 rows: 0	Pass
SV-18	Logout ends the session	Dashboard redirects to login	HTTP 302 to /login	Pass
SV-19	POST without CSRF token	Request rejected (400)	HTTP 400	Pass
SV-20	Login with wrong password	Error message, stays on login	HTTP 200	Pass

4. TC-UX-01 (core actions within 3 clicks)

Counted from the templates and nav links, starting from the dashboard after login:

Action	Clicks	Path	Result
Book and check in	2	click an available slot, click Book & Check-in	Pass
Check out	3	My Bookings, Check-out button, confirm prompt	Pass
View history	1	My Bookings	Pass
Admin add slot	2	Admin Slots, Add slot	Pass

This count was done from the page structure, not by someone using a browser, and check-out is 3 clicks only because of the confirmation prompt (deviation 3 in docs/deviations.md; issue-14 report says 2). Please do a quick manual click through in the browser before submitting if the evaluator expects it to be a manual check.

5. Not covered

Visual layout and browser rendering were not checked, and HTTPS and production deployment are out of scope (Test Plan section 5.1 and deviations table 4).